



**A Multi-Parametric Deep Learning Approach for Short-Term Load
Forecasting and Grid Anomaly Characterisation in Smart Grids**

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Abstract

Accurate short-term load forecasting (STLF) is critical for safe and efficient operation of modern electrical distribution networks, especially in rapidly growing regions where infrastructure is under increasing stress. However, existing research on data-driven forecasting and anomaly characterisation has primarily focused on North American and European power systems, leaving a gap in understanding of South Asian grid environments, which face fundamentally different operational challenges.

This project details the development and evaluation of a multi-parametric Long Short-Term Memory (LSTM) network for STLF, using two years of hourly transformer-level data from Gopalganj University of Science and Technology, Bangladesh. Eighteen features spanning electrical, meteorological, and temporal domains were engineered and systematically tested to identify the optimal input configuration. The final stacked LSTM (64-32 units) achieved a test-set RMSE of 35.23 kW, MAE of 25.13 kW, and MAPE of 8.87%. While the model narrowly underperformed a naive persistence baseline (8.83% MAPE) overall, it outperformed persistence during steady-state operation (7.65% vs 9.66% MAPE). Moreover, SHAP analysis confirmed the model's reliance on physically meaningful drivers, including phase currents, cyclical time encodings, and autoregressive lags. It revealed regime-dependent behaviour of features between peak and off-peak hours. Data-quality anomalies, such as voltage sag events and phase imbalance between currents, are characterized as diagnostic indicators that support STLF integrity. The dataset addresses a gap in South Asian transformer-level load forecasting literature and demonstrates that model transparency is achievable alongside production-relevant accuracy.

Contents

Abstract	i
List of Abbreviations	iv
List of Symbols	v
Chapter 1 Introduction	I
Chapter 2 Literature Review.....	III
2.1 Smart Grid Operations and Forecasting.....	III
2.2 Deep Learning Approaches and LSTM Networks	III
2.3 Multi-Parametric Input Representations	III
2.4 Model Interpretability & SHAP	IV
2.5 Grid Anomaly Characterisation and Data-Quality Integrity.....	IV
2.6 Research Gap	IV
Chapter 3 The Dataset.....	VI
3.1 Electrical Measurements.....	VI
3.2 Meteorological Features	VI
3.3 Engineered Features	VII
3.3.1 Active Power	VII
3.3.2 Phase Imbalance	VII
3.3.3 Cooling Degree Hours	VII
3.4 Data Cleaning	VIII
3.5 Dataset Summary.....	VIII
Chapter 4 The LSTM Network.....	X
4.1 Preprocessing.....	X
4.2 Model Architecture	X
4.3 Evaluation.....	X
Chapter 5 Results and Discussion.....	XIII
5.1 Comparative Evaluation	XIII
5.2 Feature Search Performance	XIV
Chapter 6 SHAP Analysis.....	XV

Chapter 7 Conclusion..... XVII

Reference list..... XVIII

List of Abbreviations

Abbreviation	Meaning
A	Ampere
Adam	Adaptive Moment Estimation
ARIMA	AutoRegressive Integrated Moving Average
CDH	Cooling Degree Hours
EN	European Norm
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
kW	kilowatt
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MSE	Mean Squared Error
pf	power factor
RMSE	Root Mean Squared Error
SARIMA	Seasonal AutoRegressive Integrated Moving Average
SHAP	SHapley Additive exPlanations
STLF	Short-Term Load Forecasting
V	Volt (unit of voltage)
WHO	World Health Organization

List of Symbols

Symbol	Description
CDH	Cooling degree hours
I_{mean}	Arithmetic mean of the three phase currents
I_R, I_Y, I_B	Phase currents on the Red, Yellow, and Blue phases
max()	Maximum function
N	Number of observations in a given segment/sample
P_{act}	Active power (kW)
pf	Power factor
t	Time step / hour index
T_{ambient}	Ambient temperature
V_{sys}	System voltage
°C	Degrees Celsius
$\mu(\cdot)$	Arithmetic mean (used in Phase Imbalance calculation)
$\sigma(\cdot)$	Standard deviation (used in Phase Imbalance calculation)
$\sqrt{3}$	Square root of 3

Chapter 1 Introduction

Reliable operation of modern electrical distribution networks depends on the ability to forecast near-term electricity demand accurately and to identify abnormal grid behaviours, such as voltage excursions, current imbalances, and data-quality artifacts that threaten forecast reliability. In environments facing pressure from population growth, industrial expansion, and aging infrastructure, failure in either forecasting or anomaly detection can have serious consequences, ranging from unplanned outages to errors in supply planning that risk over- or under-provisioning of generation capacity (Dong et al., 2025).

Traditional short-term load forecasting methods, including ARIMA and SARIMA, assume linearity and stationarity in time series data. These assumptions rarely hold for real-world electricity demand, which reflects non-linear relationships among temperature, humidity, time of day, day of week, and seasonal cycles. Deep learning models, particularly LSTM networks, have emerged as strong alternatives due to their ability to model complex temporal dependencies over long historical sequences, without needing explicit manual engineering of lag features (Muzaffar and Afshari, 2019; Dong et al., 2025).

Ensuring data quality remains a parallel challenge. Sensor drift, metering artifacts, and voltage or current excursions can degrade the quality of input data for forecasting. Most operational environments lack labelled fault data, making supervised classification impractical. Unsupervised and statistically driven flagging methods, profiling normal behaviour and detecting deviations via residual analysis or statistical thresholds, offers practical benefits without requiring labeled examples of abnormal events (Guato Burgos et al., 2024; Patrizi et al., 2024).

Most published research draws on North American and European grid data, which differ fundamentally from the consumption profiles, infrastructure, and operational realities of South and Southeast Asian distribution networks. Transformer-level substation data from Bangladesh remains rare in the literature. Although recent studies have begun to examine national-level demand (Hossain et al., 2025), there is little availability of transformer-level, sub-hourly, multi-parametric electrical data.

This study addresses that gap by developing a multi-parametric deep learning framework applied to real-world transformer data from the Gopalganj University of Science and Technology substation in Bangladesh. The dataset, spanning August 2023 to September 2025, combines hourly electrical measurements with co-located meteorological data. By building a unified feature space that captures the transformer's complete electrical state, the model is able to learn from interactions among load, power quality, and environmental conditions, rather than relying solely on aggregate consumption data.

The primary objective is to develop and evaluate an LSTM network for short-term active power forecasting, with performance measured by RMSE and MAPE and compared to classical benchmarks and recent

literature. A secondary objective is to apply SHAP for parameter-level interpretability, identifying which electrical and weather variables most strongly influence forecasts.

The remainder of this study is structured as follows: Section 2 reviews literature, Sections 3 and 4 present the methodology and model design, Sections 5 and 6 evaluate results, and Section 7 concludes.

Chapter 2 Literature Review

2.1 Smart Grid Operations and Forecasting

The shift from conventional electrical networks to smart grid architectures has fundamentally raised the performance standards for utility providers. Smart grids combine advanced metering, real-time sensing, and digital communications, enabling more responsive and efficient management of electricity supply and demand. This greater operational complexity makes accurate short-term load forecasting especially critical. Grid operators must anticipate load changes with precision to schedule generation, manage peaks, and avoid the costs of over-provisioning. Even modest forecasting errors can lead to financial penalties, grid instability, and outright supply failures.

Short-term load forecasting focuses on predicting electricity demand from one hour up to several days ahead. At the transformer and substation level, accurate STLF is essential to keep transformer loading within safe limits and prevent accelerated insulation degradation, thermal overload, or equipment failure. The challenge is compounded by the non-linear, non-stationary nature of electrical demand, which is shaped by temporal patterns, daily cycles, weekday-weekend differences, seasonal shifts, and changing climate conditions (Dong et al., 2025).

2.2 Deep Learning Approaches and LSTM Networks

The limitations of classical methods have driven growing interest in deep learning for load forecasting. Long short-term memory networks, in particular, have become the leading approach for time series prediction because their architecture addresses the vanishing gradient problem that limits traditional recurrent neural networks. LSTMs maintain a cell state and use learned input, forget, and output gates, allowing selective retention or discarding of information across long sequences. This is well suited to electricity demand data, which shows strong similarity at daily and weekly gaps.

LSTM models have demonstrated practical advantages over conventional techniques. For example Muzaffar and Afshari (2019) showed that adding meteorological data, such as temperature and humidity, further improved accuracy, with MAPE dropping below 5% in certain cases. More recent studies, including Dakheel and Çevik (2025), reported LSTMs achieving 1.30% MAPE on high-resolution Belgian transmission data. These results demonstrate that sub-2% MAPE is possible on high-resolution, low-noise data. However, at the transformer level, where load volatility and metering noise are greater, single-digit MAPE adds value, consistent with the outcomes observed here.

2.3 Multi-Parametric Input Representations

A consistent finding in the literature is that richer input representations lead to more robust forecasts. While most approaches rely only on historical consumption, recent work argues for including the full electrical state, voltage, phase currents, and power factor, as these variables contain predictive information

that aggregate consumption data does not. Patrizi et al. (2024) showed that multi-parametric smart meter data, including voltage and current, can reveal load signatures missed by models using only energy totals. At the transformer level, phase current imbalance and voltage variation under changing load conditions provide insight into how load is distributed across feeders and help predict future demand patterns.

Meteorological features are equally important. Temperature is the primary external driver of electricity demand in most settings, with humidity modulating cooling loads and human comfort. In tropical environments like Bangladesh, where temperatures regularly exceed 35°C and humidity nears 100% during monsoon, the interaction between temperature and humidity creates complex, non-linear demand that classical models cannot capture.

2.4 Model Interpretability & SHAP

A major limitation of deep learning forecasting models is their lack of interpretability. An LSTM network that produces accurate predictions but cannot explain which inputs are driving those predictions is difficult for grid operators to trust or act on. This has led to rising interest in post-hoc explanatory methods that break down predictions by input contribution without changing the underlying model.

The SHAP (SHapley Additive exPlanations) framework, introduced by Lundberg and Lee (2017), provides consistent, locally accurate attributions for each feature. Baur et al. (2024) found SHAP to be the most widely used post-hoc method for electric load forecasting, though most studies report only global feature importance and rarely explore explanations across different operational regimes. Applying SHAP to LSTM outputs makes it possible to identify which electrical and meteorological variables such as voltage, power factor or temperature drive demand predictions, turning a black-box model into an interpretable tool for operational support.

2.5 Grid Anomaly Characterisation and Data-Quality Integrity

Alongside forecasting, data-quality assurance is critical. The literature on anomaly detection in smart grid telemetry focuses on ensuring that input data reflects real grid states and not sensor faults or artifacts. Guato Burgos et al. (2024) identified unsupervised, threshold-based, and residual-based methods as effective when labelled fault data are unavailable. Patrizi et al. (2024) applied anomaly detection to power-quality data from smart meters, providing a methodological basis for the voltage-sag and phase-imbalance flagging. The engineered flags characterised the prevalence of non-standard operating conditions in the dataset as an exploratory diagnostic.

2.6 Research Gap

Despite extensive research on LSTM-based load forecasting and the growing recognition of SHAP for interpretability, two important gaps remain. First, most benchmarks use North American and European data, which differ fundamentally from South and Southeast Asian grids in consumption patterns,

infrastructure, and climate. Second, transformer-level, sub-hourly data from Bangladesh are rare, even as national-level studies begin to appear (Hossain et al., 2025). Bangladesh's tropical climate, with extreme temperature-humidity combinations, monsoon rainfall, and high solar radiation, creates complex demand patterns. This study addresses both gaps by applying a multi-parametric LSTM forecasting framework with SHAP interpretability. The work aims to provide a transparent forecasting methodology applicable to similar distribution networks.

Chapter 3 The Dataset

The dataset used in this study was obtained from the Gopalganj University of Science and Technology, Bangladesh. It ranges from August 2023 to September 2025, providing approximately two years of hourly observations. This is enough to capture multiple seasonal cycles, including all season behaviour of Bangladesh’s climate, as well as the university’s academic calendar, encompassing term-time periods of elevated demand and vacation periods of reduced load. It is also enriched with weather data recorded with the same frequency.

3.1 Electrical Measurements

The core electrical dataset contains five raw measurement columns recorded at hourly intervals across the two-year window, having 19,008 records before cleaning: system voltage, three independent phase currents, and power factor. The three-phase representation provides an advantage over datasets that record only aggregate consumption. Moreover, phase currents enable computation of phase imbalance indicators and give the model a richer representation of load distribution along the feeder.

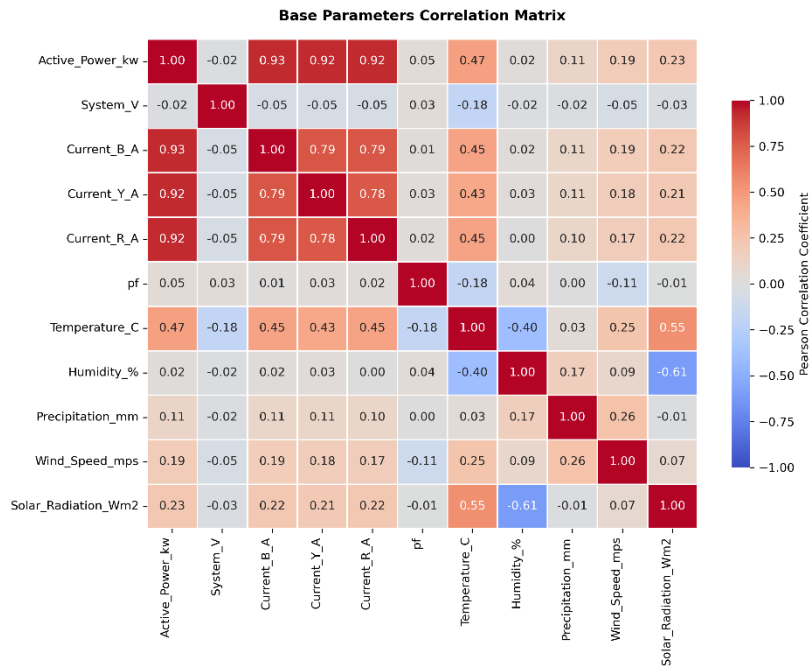


Figure 1: Correlation Matrix for dataset parameters

3.2 Meteorological Features

Temperature is the dominant driver of electricity demand in tropical climates, acting mainly through air conditioning and space cooling loads, spanning 8.8°C to 40.8°C. Relative humidity, precipitation, wind speed, and solar radiation are also recorded as referenced in Figure 1. While solar radiation and temperature capture daily and seasonal behaviour most directly and are retained in the final feature set, humidity, precipitation, and wind speed were tested and found not to improve forecasting and hence dropped.

3.3 Engineered Features

Beyond the raw measurements, several engineered features were derived in an effort to improve model training and performance.

3.3.1 Active Power

The target variable, active power in kilowatts, was computed using the standard three-phase power formula:

$$P_{\text{act}} \text{ (kW)} = \frac{\sqrt{3} \times V_{\text{sys}} \times I_{\text{mean}} \times \text{pf}}{1000}$$

where I_{mean} represents the arithmetic mean of the currents.

Active power is the actual electrical energy consumed to perform work. Accurate forecasting is essential for maintaining a balance between power generation and consumer demand.

3.3.2 Phase Imbalance

In electrical networks, phase imbalance occurs when currents across three phases of a system are not equally synchronised. Significant deviations from balanced phases can indicate grid stress or potential equipment failure.

Being a crucial grid data quality and system stress parameter, phase imbalance percentage was calculated and kept for its significance, using:

$$\text{Phase Imbalance (\%)} = \frac{\sigma(I_R, I_Y, I_B)}{\mu(I_R, I_Y, I_B)} \times 100$$

3.3.3 Cooling Degree Hours

A feature, Cooling Degree Hours (CDH), was created to capture non-linear relationship between ambient temperature and HVAC-driven electricity consumption.

$$CDH_t = \max(0, T_{\text{ambient},t} - 24^\circ\text{C})$$

where T_{ambient} is the ambient temperature at hour t .

The World Health Organization (WHO, 2018) identifies 24°C as a relevant threshold for assessing potential health effects associated with elevated indoor temperatures. This threshold was therefore adopted for the CDH calculation to represent increasing cooling demand above 24°C.

This feature was one of seven candidates tested during systematic feature testing and was retained in the final model, improving test-set MAPE from 8.95% to 8.87%.

3.4 Data Cleaning

Assessing the dataset revealed recorded information to be well within defined industrial standards for electricity consumption. However, physical bounds clipping was applied to current measurements only to remove sensor errors and metering artifacts. Phase currents were processed in two stages: a hard upper cap of 1,500A to remove physically impossible readings (Figure 2), followed by a statistical IQR-based clip computed on the post-capped distribution.

System voltage was already within the tolerance brackets around the 400V nominal according to IEC 60038 and EN 50160. Similarly, Power factor also complied with IEEE Standard 141 operational guidance.

Significant voltage-drop events, denoted as Voltage Sag, were identified using a regression-based approach. A linear regression was fitted between mean phase current and system voltage, and observations where voltage fell more than two standard deviations below the regression prediction while simultaneously carrying current above the 60th percentile were flagged. This flagging characterises abnormal grid states. Distinct from routine load-dependent voltage sag, it is reported here as a grid anomaly characterisation output rather than a supervised classification target.

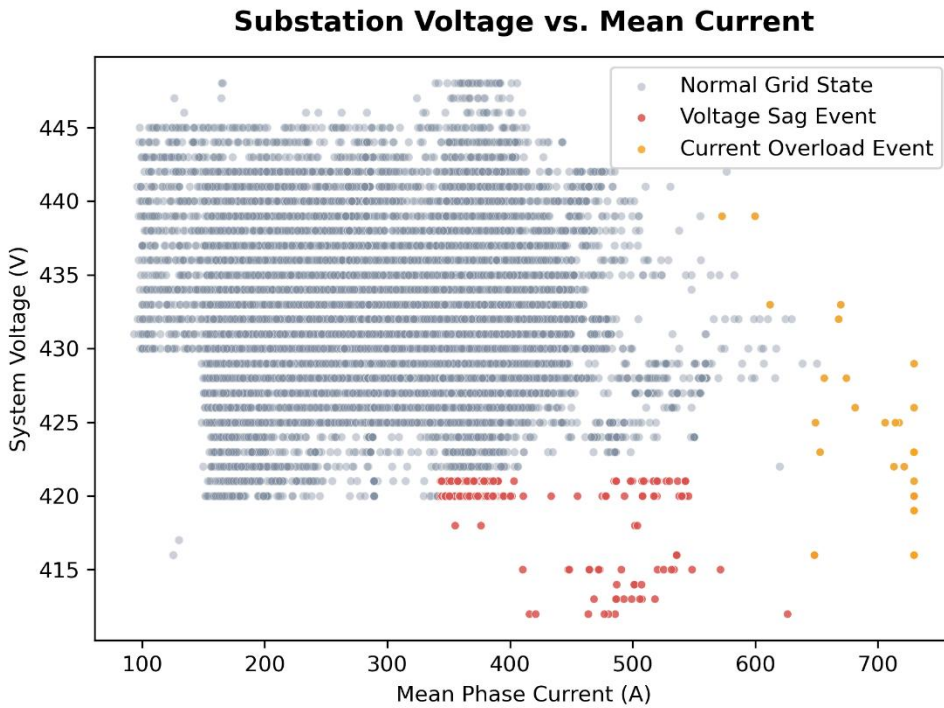


Figure 2: Spread of Anomalous Events between Mean Current and Voltage

3.5 Dataset Summary

Following cleaning and feature engineering, the final dataset comprised 19,008 hourly observations spanning August 2023 to September 2025 and 19 features. All corrections were applied via clipping and imputation; some temporarily created features were deleted. Preserving full temporal continuity is essential for sequence-based model training. Hence, the dataset was partitioned chronologically into training (70%, 8 August 2023 – 7 February 2025, 13,188 observations), validation (15%, 7 February 2025 – 5

June 2025, 2,826 observations), and test (15%, 5 June 2025 – 30 September 2025, 2,826 observations) subsets. Figure 3 shows that the test partition exhibits higher volatility than the earlier training data, resulting in a data distribution drift problem. This directly explains the regime-dependent performance gap between LSTM and persistence discussed later.

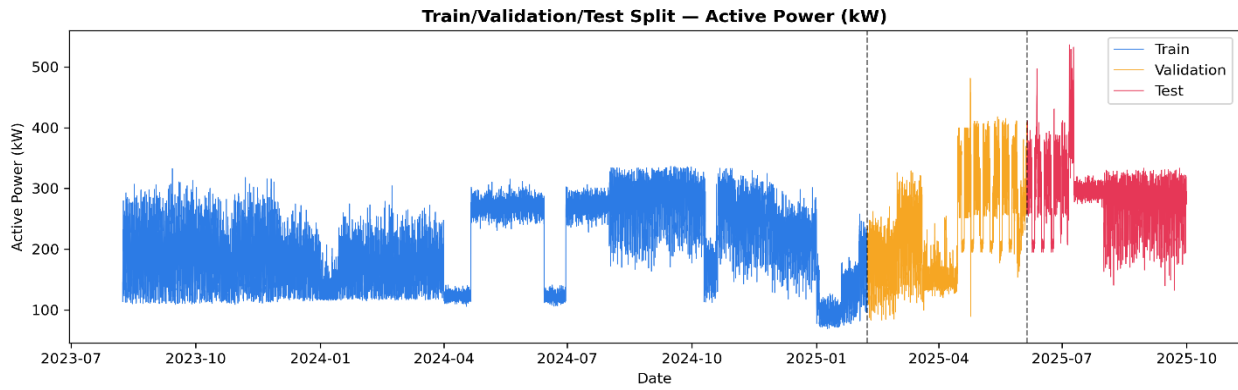


Figure 3: Dataset Split

Analysing the autocorrelation function plot (Figure 4) over a 72-hour period revealed repetitive peaks at the 24-, 48-, and 72-hour marks, confirming seasonal usage in the substation load. These findings justified the inclusion of 24h, 48h, and 168h lag features to capture autoregressive dependencies.

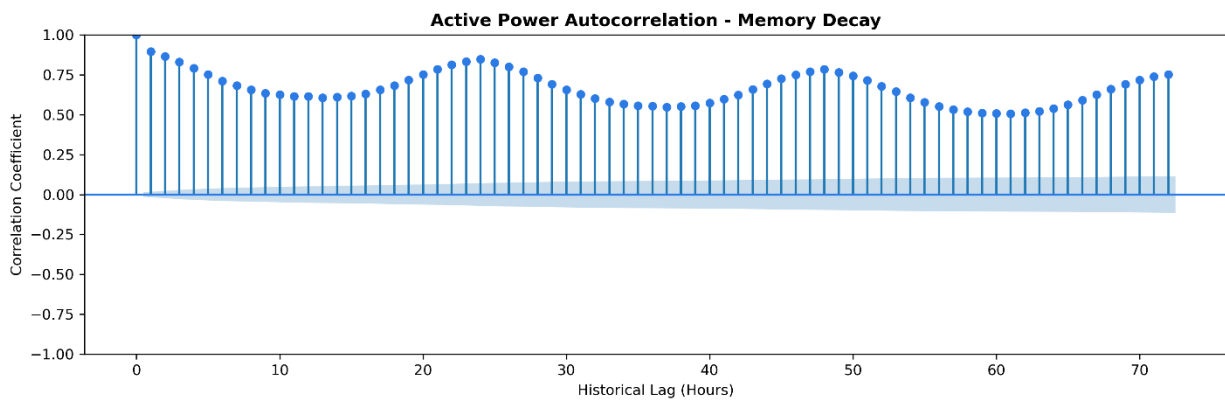


Figure 4: Historical Correlation of Active Power

Chapter 4 The LSTM Network

4.1 Preprocessing

Along with the historical lag steps, generated for active power and ambient temperature at 24, 48, and 168 hours before the current time step, periodic time variables were transformed into cyclic features using sine and cosine functions. This avoided the discontinuity in raw integer values for hour (e.g., 0-23) and month.

Systematically testing seven engineered features one by one led to an 18-dimensional input space, having CDH alongside the original base features: 5 electrical signals (Voltage, 3 phase currents, pf), 3 weather features (Temperature, Solar Radiation, CDH), 4 cyclical temporal features (sin and cosine hour/month), and 6 lags (power and temperature for 24/48/168). Furthermore, humidity, precipitation, wind speed, phase-imbalance percentage, and vacation flag were tested and pruned as they did not improve validation accuracy.

A manual chronological split was used instead of random or Time Series Split cross-validation, preserving sequential dependencies and mirroring real-world implementation (train on history, evaluate on future). All features and the target were Min-Max scaled to $[0,1]$ and fitted exclusively on the training partition to prevent leakage, with separate scalers for features and target, allowing exact inverse transformation. A sliding-window transformation converted the scaled matrices into 3D shape (samples, 48 timesteps, 18 features) for LSTM.

4.2 Model Architecture

A stacked two-layer LSTM was implemented. A primary layer of 64 units with full sequence returns, feeding a second layer of 32 units compressing time series dynamics into a single state vector, with 20% dropout after each block. The model was compiled with Adam (learning rate and gradient-norm clipping) and trained to minimise Mean Squared Error, penalising extreme peak-demand deviations. A three-tier callback approach was used to govern training: ReduceLROnPlateau (patience 7, lowering the learning rate on stagnant validation loss), ModelCheckpoint (persisting the best validation-loss weights), and EarlyStopping (patience 15, restoring best weights), together allowing two learning-rate reductions before termination, balancing convergence depth against overfitting risk. Training was set to a maximum of 100 epochs, with a batch size of 64 and shuffling disabled to preserve temporal continuity across mini-batches.

4.3 Evaluation

The optimised LSTM achieved a test-set RMSE of 35.23 kW, MAE of 25.13 kW, and MAPE of 8.87%, as shown in Table 1. While this single-digit MAPE meets typical operational requirements for short-term load forecasting, it is higher than the sub-2% MAPE reported by Dakheel and Çevik (2025) for high-resolution transmission-level data. This difference is likely due to the hourly resolution and greater load variability present in datasets like the one used in this study.

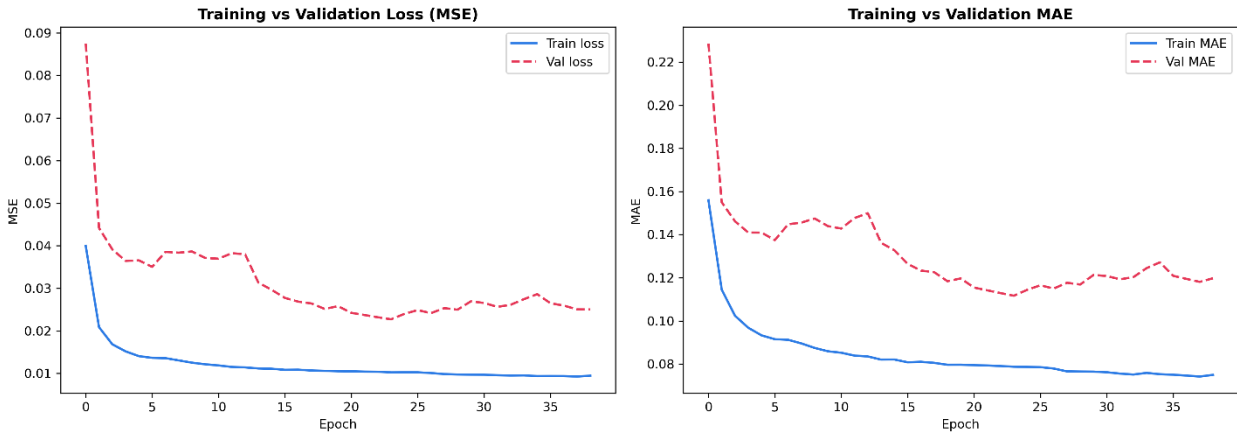


Figure 5: Training and Validation Curves for the Model

As shown in Figure 5, validation loss reached its minimum around epoch 24 (MSE ≈ 0.023), and the narrow, stable gap between training and validation curves confirms dropout and gradient thresholding effectively controlled overfitting across the feature space.

However, the model tends to underestimate peak load surges (≥ 450 kW) and smooth out minimum night-time troughs (Figure 6), a common effect of MSE optimisation, which favours average values over extreme spikes. Residual analysis indicates a mean bias of +3.15 kW (less than 1.1% of the mean load), with no significant overall calibration error. Yet, consistent under-forecasting of peaks is evident, with right-skewed residuals (maximum under-prediction +182.57 kW, maximum over-prediction -114.32 kW).

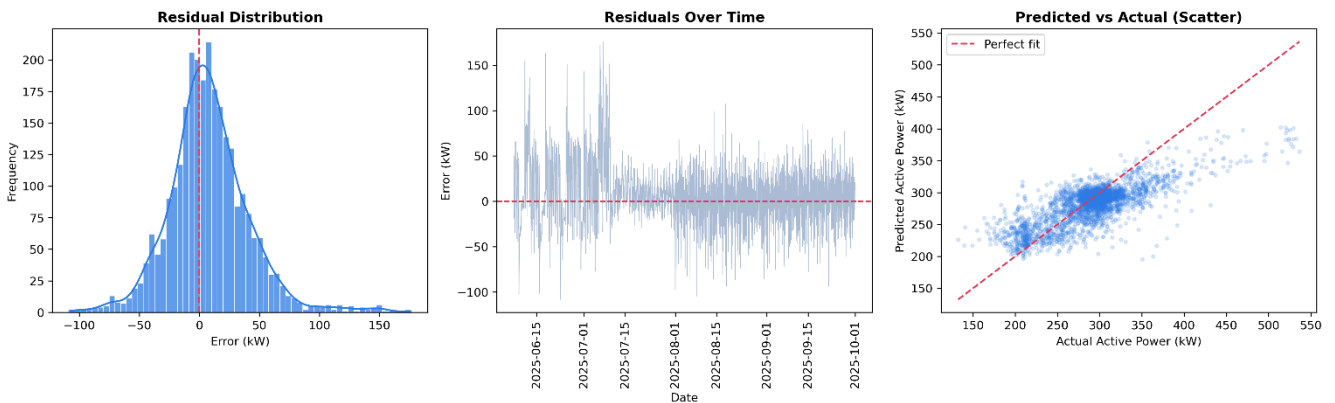


Figure 6: Residual Plots Assessing Model Performance

Furthermore, as illustrated in Figure 7, the model follows the actual pattern closely in calmer periods but lags 1–2 hours behind transient spikes during volatile intervals (e.g., the 2025-06-12 peak near 500 kW), highlighting regime-dependent performance. This suggests that an operational reserve margin may be required to mitigate peak underestimation.

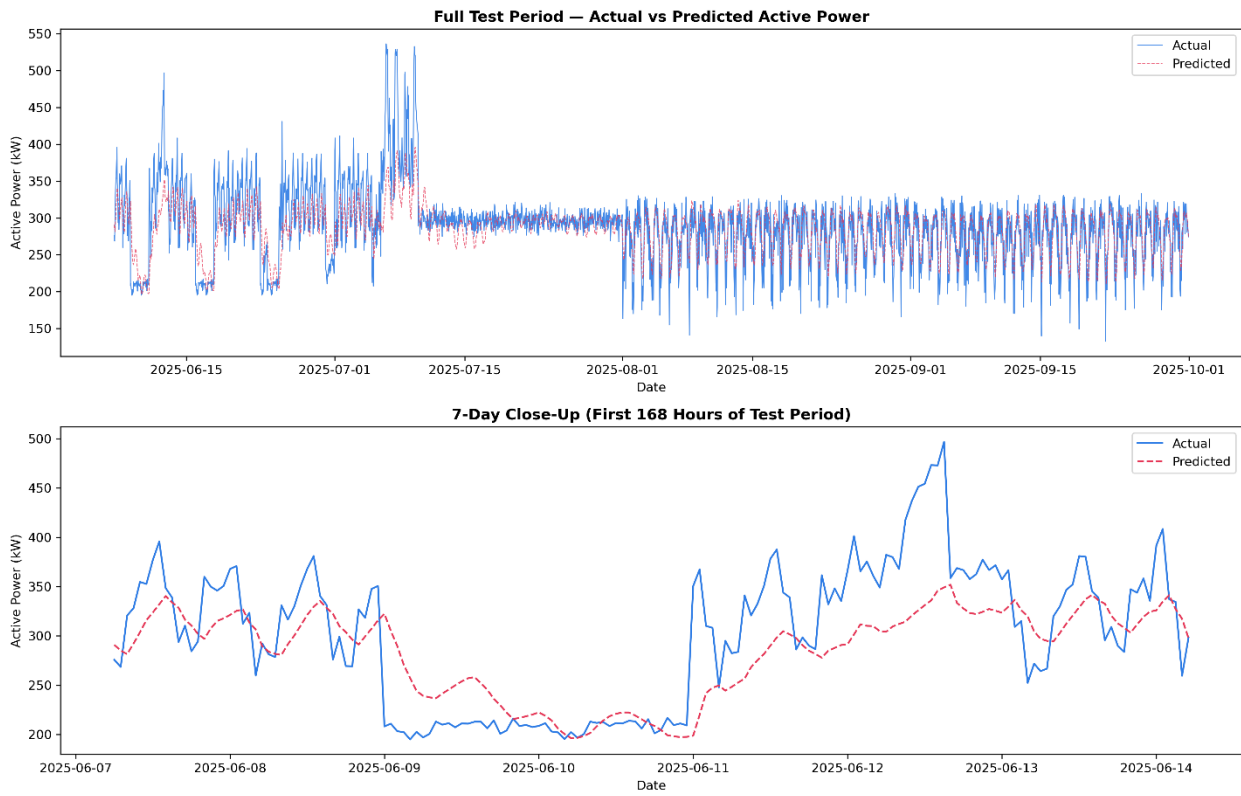


Figure 7: Predicted Active Power Plots

Chapter 5 Results and Discussion

5.1 Comparative Evaluation

The LSTM model’s predictions were compared with a naive persistence baseline, with performance evaluated on both the overall test set and within specific operational modes.

Model	RMSE (kW)	MAE (kW)	MAPE (%)
Persistence (Naive Baseline)	33.68	24.18	8.83%
Stacked LSTM (Proposed)	35.23	25.13	8.87%

Table 1: Global performance comparison

Operational Segment	n	LSTM MAPE	Persistence MAPE	Dominant Model
Calm Window ($\geq 2025-07-14$)	1,896	7.65%	9.66%	LSTM (+2.01%)
Volatile Window ($< 2025-07-14$)	882	11.50%	7.04%	Persistence (+4.46%)
Term-Time	2,568	8.65%	9.01%	LSTM (+0.36%)
Vacation Period	210	11.53%	6.64%	Persistence (+4.89%)

Table 2: Regime-segmented performance comparison

- Steady-State Advantage:** During steady-state periods, which comprise 68% of the test set, the LSTM model demonstrated a clear advantage over the persistence baseline, achieving a lower MAPE (7.65% vs. 9.66%). This result shows that the model can recognize daily and time-based patterns found in normal operations.
- Transient Volatility:** During periods of high volatility, the persistence model produces lower errors because it always uses the most recent value, resulting in no delay. In contrast, the LSTM’s use of multi-hour lookback and dropout regularisation causes its predictions to be smoother and slightly delayed, so it briefly lags behind sudden, sharp changes.
- Comparison to Recent Work:** This pattern echoes Al Amin Sarker et al. (2026), who note that volatility-driven error remains the dominant failure mode across grid models generally, rather than being a limitation specific to this university dataset.
- Data-Quality & Feature Limitations:** The binary vacation feature (n=210) doesn’t line up with the actual operational changepoint on 14 July 2025 (n=882). This shows that relying solely on external calendar flags isn’t enough to explain major load shifts. This is a limitation of the dataset that could be addressed with access to internal records, like facility maintenance logs or occupancy data.
- Hybrid Framework:** The difference in model performance across stable and volatile periods suggests that a single network is not well suited to handle both situations at once. Future work should explore a regime-aware approach, where a volatility or changepoint detector decides which

forecasting method to use: a short-horizon persistence model for sudden changes, and the LSTM for periods with regular, predictable patterns.

5.2 Feature Search Performance

To understand the performance measure of the final architecture, 11 configurations were additionally tested, varying the lookback window (24h, 48h, 72h), including or excluding rolling lag features, and exploring four architecture sizes, three learning rates, and two batch sizes. This ensured that the selected setup was not arbitrary, as the results show in Table 3.

Configuration Number	Configuration	MAPE	Persistence MAPE
1	arch_64_32_dr0.2 (final model)	8.87%	8.83%
2	lookback_48	10.24%	8.83%
3	lr_0.0001	10.42%	8.83%
4	lookback_24	10.98%	8.81%
5	arch_128_64_dr0.3	11.19%	8.83%
6	lookback_72	11.30%	8.84%
7	lookback_48_plus_rollstd24	11.59%	8.83%
8	arch_128_64_dr0.1	11.60%	8.83%
9	arch_256_128_dr0.2	12.08%	8.83%
10	batch_32	12.71%	8.83%
11	lr_0.001	12.81%	8.83%

Table 3: Exploratory configuration testing results

Based on this search, the following observations were made:

1. The 64/32-unit architecture delivered better performance than both the larger 128/64 and 256/128 variants, suggesting that increasing model capacity leads to over-smoothing rather than improved accuracy.
2. Extending the lookback window resulted in worse performance, with a 72-hour window resulting in a 11.30% MAPE compared to 8.87% for the 48-hour window. This indicates that adding additional historical data weakens the short-lag signal instead of providing useful context.
3. Testing these features (such as rollstd24, which gave 11.59% MAPE) caused the network to predict more of a trailing average, making it less responsive to real spikes and troughs in the data. For this reason, rolling-window features were excluded from the final model.
4. None of the tested configurations outperformed the persistence baseline on the test set. The LSTM's advantage is limited to the calm window identified in the time-segmented breakdown.

Chapter 6 SHAP Analysis

To investigate the decision-making process of the LSTM, SHAP (Gradient Explainer) values were calculated over 200 test sequences and aggregated on the same lookback window to assess overall feature importance. As shown in Figure 8, the three phase currents (Current Y ≈ 0.0017 , Current R ≈ 0.0014 , Current B ≈ 0.0017) emerged as the most influential features, followed by the power lags 48 and 24. In contrast, CDH and temperature had much lower importance.

The SHAP analysis shows a physically coherent hierarchy. Phase currents dominate because active power is derived from them, confirming the model has learned the underlying electrical relationships. The lag features at 24 and 48 reflect the repetitive behaviour, while the time encodings reflect daily and seasonal demand changes.

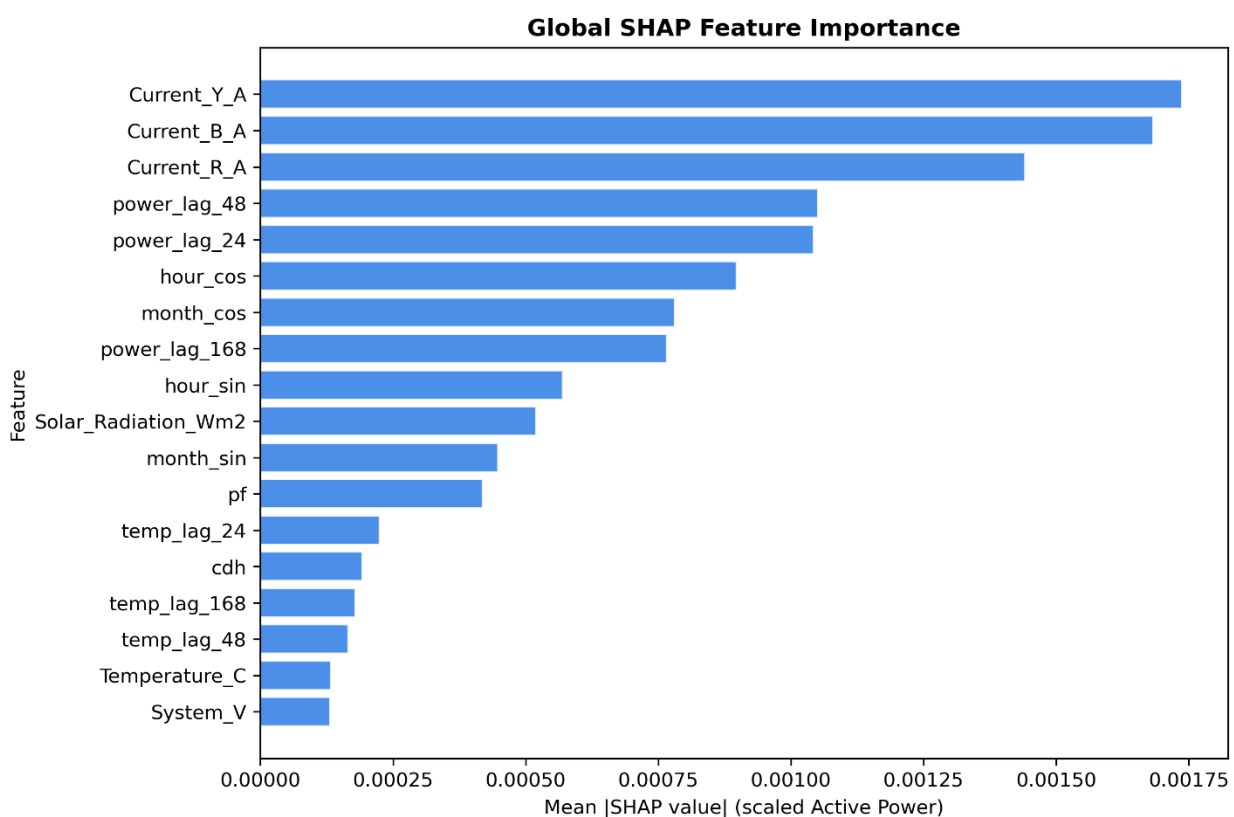


Figure 8: Importance of Each Parameter on Model Performance

Figure 9 supports this by showing that the hour feature shifts from positive during peak hours to negative off-peak. This demonstrates the model understands the natural ups and downs of daily power use, rather than assuming electricity demand changes in a straight line throughout the day. Overall, these results confirm that the LSTM is making decisions grounded in real-world patterns, recognizing that time of day can increase or decrease power demand.

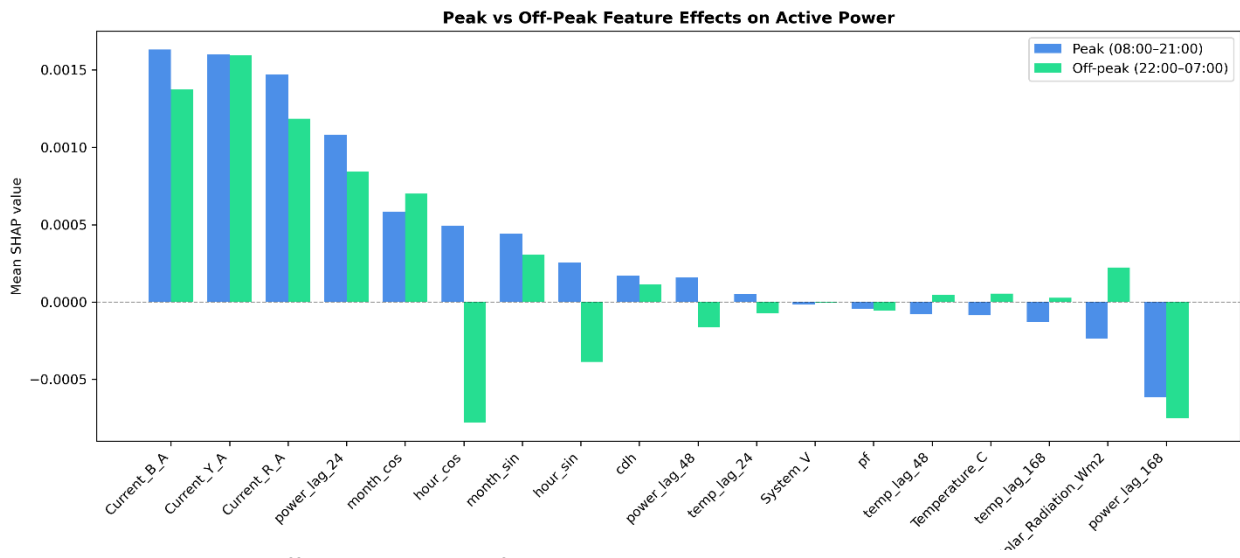


Figure 9: Peak vs Off-Peak Behaviour of Parameters

Chapter 7 Conclusion

This study aimed to develop an interpretable, multi-parametric LSTM forecaster for a data-scarce grid environment, and to characterise data quality anomalies. The model achieved a test MAPE of 8.87%, narrowly trailing naive persistence overall (8.83%), but showing a substantial advantage in steady-state operation and underperforming during abrupt transients. This regime-dependent pattern emerges as the project's most actionable finding, highlighting the need for a hybrid architecture. SHAP analysis confirms that the model's forecasts rely on features that have a clear physical basis, such as phase currents, time-of-day, and recent load history. The peak/off-peak disaggregation demonstrates that the network understands time-of-day, rather than applying a single global rule. Engineered data quality indicators offer a complementary grid characterisation layer, independent of the forecasting model. Principal limitations include the single-site dataset and the coarse vacation flag that does not reflect the actual operational changepoint.

Future work should prioritise developing a rolling-volatility changepoint detector to direct forecasts between the LSTM and a short-term persistence model and should include validation using data from additional transformer sites.

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